

Skyguide surveillance data provision as CIS

SWIM Service Description Document

General Service Description	
Service description identification SWIM-SERV-030	Title: Skyguide surveillance data provision as CIS Edition: V0.2 Publication Date: TBD
Service identification SWIM-SERV-040	Name: Provision of surveillance data as part of Skyguide ANSP CIS provision SWIM Service Version: V1.0
Service abstract SWIM-SERV-050	Skyguide's obligations in the provision of airspace conditions as part of CIS are defined in Skyguide's document "Concept of Operation / Surveillance data provision". To comply with the requirements defined in Skyguide's Concept of Operation, the service implementation is based on the ASTM F3411-22a standard (already identified as an applicable standard in U-space accepted means of compliance and used for Network Remote-ID).
Service Provider and Contact Information	
Service Provider SWIM-SERV-060	Name: Skyguide Description of organization: Swiss ANSP
Service Characteristics	
Geographical extent of information SWIM-SERV-090	Geographical coverage of exchanged information service payload is available through the DSS under ASTM Section A2. Skyguide will provide data within at least the Swiss border . The DSS will provide discovery (based on the volume of interest) to the various uss-base-url of the relevant areas defined (expected to be large grids of around 20NM). Skyguide individual grid end-points will close connection for requests nor overlapping with them. Skyguide reserves the right to change the shape of the areas over time, so spread the load more evenly, if it becomes necessary. DSS discovery is therefore mandatory.

	Note: for the Zurich soft launch, one single grid will be centered on the USpace airspace.
Service categories SWIM-SERV-100	Service category: Surveillance Data
Service lifecycle information SWIM-SERV-110	Lifecycle stage and if applicable qualifying dates: Until further notice
Service standard reference SWIM-SERV-120	Service standard reference: 1. Service Standard Reference: Standard Specification for Remote ID and Tracking (ASTM F3411-22a) Implemented options of the service standard: • Discovery and publication of surveillance data is implemented, using ASTM F3411-22a ○ The DSS discovery mechanism, namely the endpoint to retrieve active areas in the for of USSBaseURL: GET /dss/identification_service_areas ○ The provision of flights using the endpoint GET /uss/flights ■ The response, when the query volume is successfully overlapping with the area served on this USSBaseURL) triggers a SSE (server side event) stream of data ■ The stream of data provides event (one event per ADS-B message) using the same data structure as the response of the standard response of the API, with an additional “source” field containing the raw ADS-B message. Skyguide has several (5 in 2026) multilateration systems, each with 7 to 36 antennas. In this setup, the same ADSB-message could be provided up to 5 times (aggregated in each of the multilateration system), although highly unlikely at low altitudes. The following sample shows such an event. <pre>{ "id": { "icao": "491d83" }, "aircraft_type": "NotDeclared", "current_state": { "timestamp": { "format": "RFC3339", "value": "2026-02-27T16:04:05.796875Z" }, "position": { "lat": 46.193506233394146, "lng": 6.058753933757544, "alt": 3200.4 }, "track": 0, </pre>

	<pre> "speed": 0, "vertical_speed": 0 }, "source": { "id": "skyguide/cat21", "format": "cat21", "data": "FQBa8x/zW90jBAABAAxVARBsorYCJ3szSR2DcPrMJgDs2nD6gQBCf2sGkDH3FbASAZAAf/YF qH5WcPrmOKFy3wLgA8GRABDFk8HQAwmGAakGBhEHxAhQBAAA" } } </pre> Skyguide will provide a DSS. Provision of a DSS by the USSP is not required. However it remains possible and is described in discovery and synchronization service configuration process - as defined by InterUSS DSS Pooling instructions on joining an existing pool with new instance.
High-level Description of Service Offer	
Operational environment SWIM-SERV-130	Refer to the following documents for operational needs addressed by the service and the capabilities offered by the service. • Skyguide's Concept of Operation • Commission Implementing Regulation (EU) 2021/664 - Preamble • ASTM F3411-22a Section A2 capabilities include: - USS-DSS Interfaces - USS-USS Interfaces (proposed update with raw ADS-B information and SSE response)
Service functions SWIM-SERV-140	Refer to the following documents for functions offered by the service and their real world effects. • Skyguide's Concept of Operation • Commission Implementing Regulation (EU) 2021/664 Article 5 • ASTM F3411-22a Section A2 capabilities include: - USS-DSS Interfaces - USS-USS Interfaces (proposed update with raw ADS-B information and SSE response)
Limitations and Constraints on Using the Service	
Service access and use conditions SWIM-SERV-150	Service access and use conditions are prescribed by FOCA, as required by Regulation (EU) 2021/664, Art. 3 (4), as part of the U-space services performance requirements.
Security constraints SWIM-SERV-160	ASTM F3411-22a Sections 5.5.2.4 and A2 (relevant requirements are listed below): * NET0210 * NET0220

	* NET0310 * DSS0020 <i>Authentication is ensured by the oauth2 setup used for Network Remote ID in Switzerland</i>
Quality Aspects	
Quality of service SWIM-SERV-180	Skyguide's Concept of Operation defines various data and service quality requirements. In regards to service availability, the service is provided with a 99% guarantied availability.
Service validation information SWIM-SERV-200	Service will be verified using the InterUSS Automated Test Baseline prior to software deployment using test baseline identifier XXXX.
Behavior of the Service	
Application message exchange pattern SWIM-SERV-210	Refer to the following sections in ASTM F3411-22a for information about the application message exchange pattern(s) used by the service. * A2 - Interoperability requirements and DSS testing * A3 – https://github.com/skyguide-ansp/cis-interfaces/tree/main/surveillance
Service behavior SWIM-SERV-220	Service behaves as a service provider in F3411-22a and a basic DSS discovery service. However certain limitations have been amended from table A3.4, namely : - NetMaxDisplayAreaDiagonal will not be checked by skyguide, as a service provider. An area of any size is accepted. - ADS-B messages are provided as received from the skyguide CNS systems, typically at 1Hz, if so received. In order to address degradations, the service will : - Provide offline dual coverage to USSPs, so they can assess the coverage for their operation - Stop provision of the service when more than one ADS-B antenna is unserviceable, ensuring that the nominal coverage is always available in a redundant manner. This means that the connectivity (SSE stream) will be closed by the server (skyguide), which will not allow new connections until restored. Note that dual failure for multilateration systems causes skyguide 24/7 technicians to immediately fix the system.

	USSPs will be informed of scheduled maintenance of skyguide systems in advance as external stakeholders in skyguide's WAC process (Work on ATM Component, internal skyguide process).
Service Implementation and Structural Details	
Service interfaces SWIM-SERV-240	Refer to the following sections in ASTM F3411-22a for information about service interfaces. * A2 - Interoperability requirements and DSS testing * A3 – https://github.com/skyguide-ansp/cis-interfaces/tree/main/surveillance
Service interface protocols and data format SWIM-SERV-260	Refer to the following sections in ASTM F3411-22a for information about service interface protocols and payload. * A2 - Interoperability requirements and DSS testing * A3 – https://github.com/skyguide-ansp/cis-interfaces/tree/main/surveillance <u>Note:</u> Altitude provided in the F3411-22a envelope is the same altitude as in the embedded raw ADS-B message. It is therefore subject to the same limitations, namely accuracy and origin (Barometric and therefore not QNH corrected or Geometric in AMSL) Similarly, horizontal and time information in the envelope are also subject to the accuracy specified inside the raw ADS-B message itself.
Service operations SWIM-SERV-270	Refer to the following sections in ASTM F3411-22a for information about service operations. * A2 - Interoperability requirements and DSS testing * A3 – https://github.com/skyguide-ansp/cis-interfaces/tree/main/surveillance
Service messages SWIM-SERV-280	Refer to the following sections in ASTM F3411-22a for information about messages exchanged by the service. * A2 - Interoperability requirements and DSS testing * A3 – https://github.com/skyguide-ansp/cis-interfaces/tree/main/surveillance
Information Aspects of the Service	
Information definition (minimum)	Refer to the following sections in ASTM F3411-22a for information about the exchanged information service payload. * A2 - Interoperability requirements and DSS testing * A3 – https://github.com/skyguide-ansp/cis-interfaces/tree/main/surveillance

SWIM-SERV-290	interfaces/tree/main/surveillance
Resources	
Machine-readable service interface definition SWIM-SERV-320	Refer to the following sections in ASTM F3411-22a for information about the exchanged information service payload. * A2 - Interoperability requirements and DSS testing * A3 – https://github.com/skyguide-ansp/cis-interfaces/tree/main/surveillance
Abbreviations and acronyms SWIM-SERV-350	Refer to the following sections in ASTM F3411-22a for abbreviations and acronyms: * 3 - Terminology